Materia:

Pruebas de Software y Aseguramiento de la Calidad

Actividad:

Ejercicio de programación 3 y pruebas de unidad

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Fecha de entrega:

22 de febrero de 2026

Problemas

1. Reservation System

Source

```
"""Reservation System Module

CLA-T-1201 - A01796592 - Assignment 6.2 - Problem 1

This module contains the ReservationSystem class which manages reservations.

"""


import json

import os


class Hotel:

    """Hotel class to represent a hotel with its name and location."""

    def __init__(self, hotel_id):

        self.hotel_id = hotel_id


    def create_hotel(self, name, location, rooms, reservations=None):

        """Creates a hotel instance and returns it."""

        if rooms <= 0:

            raise ValueError("Number of rooms must be greater than 0.")

        hotel_data = {

            "hotel_id": self.hotel_id,

            "name": name,

            "location": location,

            "rooms": rooms,

            "reservations": reservations if reservations is not None else []

        }

        with open("hotels.json", "w", encoding="utf-8") as f:

            json.dump(hotel_data, f, indent=4)

        return hotel_data


    def delete_hotel(self):

        """Deletes the hotel instance from the hotels.json file."""

        if os.path.exists("hotels.json"):

            os.remove("hotels.json")

            print("Hotel deleted successfully.")
```

```
    else:
        print("Hotel file does not exist.")

def display_hotel_information(self):
    """Displays the hotel information."""
    if os.path.exists("hotels.json"):
        with open("hotels.json", "r", encoding="utf-8") as f:
            hotel_data = json.load(f)
            print(f"Hotel Name: {hotel_data['name']}")
            print(f"Location: {hotel_data['location']}")
            print(f"Rooms: {hotel_data['rooms']}")
            print(f"Reservations: {hotel_data['reservations']}")
    else:
        print("Hotel file does not exist.")

def modify_hotel_information(self, name=None, location=None, rooms=None):
    """Modifies the hotel information and updates the hotels.json file."""
    if os.path.exists("hotels.json"):
        with open("hotels.json", "r", encoding="utf-8") as f:
            hotel_data = json.load(f)
        if name:
            hotel_data["name"] = name
        if location:
            hotel_data["location"] = location
        if rooms:
            if rooms <= 0:
                raise ValueError("Number of rooms must be greater than 0.")
            hotel_data["rooms"] = rooms
        with open("hotels.json", "w", encoding="utf-8") as f:
            json.dump(hotel_data, f, indent=4)
    else:
        print("Hotel file does not exist.")

def reserve_room(self, customer_id, check_in_date, check_out_date):
    """Reserves a room for a customer if available."""
    if os.path.exists("hotels.json"):
        with open("hotels.json", "r", encoding="utf-8") as f:
            hotel_data = json.load(f)
```

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            reservations = hotel_data.get("reservations", [])
            rooms = hotel_data.get("rooms", 0)
            if len(reservations) < rooms:
                reservation_id = len(reservations) + 1
                reservation = {
                    "reservation_id": reservation_id,
                    "customer_id": customer_id,
                    "check_in_date": check_in_date,
                    "check_out_date": check_out_date
                }
                reservations.append(reservation)
                hotel_data["reservations"] = reservations
                with open("hotels.json", "w", encoding="utf-8") as f:
                    json.dump(hotel_data, f, indent=4)
                print(f"Room reserved successfully for customer {customer_id}")
            else:
                print("No rooms available for the selected dates.")
        else:
            print("Hotel file does not exist.")

    def cancel_reservation(self, reservation_id):
        """Cancels a reservation based on the reservation ID."""
        if os.path.exists("hotels.json"):
            with open("hotels.json", "r", encoding="utf-8") as f:
                hotel_data = json.load(f)
            reservations = hotel_data.get("reservations", [])
            reservations = [res for res in reservations
                            if res["reservation_id"] != reservation_id]
            hotel_data["reservations"] = reservations
            with open("hotels.json", "w", encoding="utf-8") as f:
                json.dump(hotel_data, f, indent=4)
            print(f"Reservation {reservation_id} cancelled successfully.")
        else:
            print("Hotel file does not exist.")

class Customer:
    """Customer class to represent a customer with their ID and name."""
    def __init__(self, customer_id):
```

```
    self.customer_id = customer_id

def create_customer(self, name, age=None):
    """Creates a customer instance and returns it."""
    if age is not None and age <= 0:
        raise ValueError("Age must be a positive integer.")
    customer_data = {
        "customer_id": self.customer_id,
        "name": name,
        "age": age
    }
    with open("customers.json", "w", encoding="utf-8") as f:
        json.dump(customer_data, f, indent=4)
    return customer_data

def delete_customer(self):
    """Deletes the customer instance from the customers.json file."""
    if os.path.exists("customers.json"):
        os.remove("customers.json")
    else:
        print("Customer file does not exist.")

def display_customer_information(self):
    """Displays the customer information."""
    if os.path.exists("customers.json"):
        with open("customers.json", "r", encoding="utf-8") as f:
            customer_data = json.load(f)
            print(f"Customer ID: {customer_data['customer_id']}")
            print(f"Name: {customer_data['name']}")
            print(f"Age: {customer_data.get('age', 'Not specified')}")
    else:
        print("Customer file does not exist.")

def modify_customer_information(self, name=None, age=None):
    """Modifies the customer information and updates
    the customers.json file."""
    if os.path.exists("customers.json"):
        with open("customers.json", "r", encoding="utf-8") as f:
```

```
            customer_data = json.load(f)
        if name:
            customer_data["name"] = name
        if age is not None:
            customer_data["age"] = age
        with open("customers.json", "w", encoding="utf-8") as f:
            json.dump(customer_data, f, indent=4)
    else:
        print("Customer file does not exist.")

class Reservation:
    """Reservation class to represent a reservation with its details."""
    def __init__(self, reservation_id, hotel_id, customer_id):
        self.reservation_id = reservation_id
        self.hotel_id = Hotel(hotel_id).hotel_id
        self.customer_id = Customer(customer_id).customer_id

    def create_reservation(self, hotel_name, check_in_date,
                           check_out_date):
        """Creates a reservation instance and returns it."""
        if check_in_date >= check_out_date:
            raise ValueError("Check-out date must be after check-in date.")
        reservation_data = {
            "reservation_id": self.reservation_id,
            "hotel_name": hotel_name,
            "hotel_id": self.hotel_id,
            "customer_id": self.customer_id,
            "check_in_date": check_in_date,
            "check_out_date": check_out_date
        }
        with open("reservations.json", "w", encoding="utf-8") as f:
            json.dump(reservation_data, f, indent=4)
        return reservation_data

    def cancel_reservation(self):
        """Cancels the reservation instance from the reservations.json file."""
        if os.path.exists("reservations.json"):
            os.remove("reservations.json")
```

```
    else:
        print("Reservation file does not exist.")
```

Test

Pylint - PEP 8

```
PS C:\Users\anton\Documents\TecMonterrey\Pruebas de Software y Aseguramiento de la Calidad\CLA-TC4017-Ene2026_PSAC\A01796592_A6.2\P1> python -m pylint .\reservation_system.py

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Your code has been rated at 10.00/10
```

Flake

```
PS C:\Users\anton\Documents\TecMonterrey\Pruebas de Software y Aseguramiento de la Calidad\CLA-TC4017-Ene2026_PSAC\A01796592_A6.2\P1> python -m flake8 .\reservation_system.py
PS C:\Users\anton\Documents\TecMonterrey\Pruebas de Software y Aseguramiento de la Calidad\CLA-TC4017-Ene2026_PSAC\A01796592_A6.2\P1>
```

Unit Test

```
PS C:\Users\anton\Documents\TecMonterrey\Pruebas de Software y Aseguramiento de la Calidad\CLA-TC4017-Ene2026_PSAC\A01796592_A6.2\test\unit\P1> python -m flake8 .\reservation_systen_test.py
PS C:\Users\anton\Documents\TecMonterrey\Pruebas de Software y Aseguramiento de la Calidad\CLA-TC4017-Ene2026_PSAC\A01796592_A6.2\test\unit\P1>
```

```
PS C:\Users\anton\Documents\TecMonterrey\Pruebas de Software y Aseguramiento de la Calidad\CLA-TC4017-Ene2026_PSAC\A01796592_A6.2\test\unit\P1> python -m pylint .\reservation_sy
sten_test.py
************* Module .\reservation_systen_test.py
.\reservation_systen_test.py:1:0: F0001: No module named .\reservation_systen_test.py (fatal)
```

Result

Coverage report: 99%

Files | Functions | Classes

coverage.py v7.13.4, created at 2026-02-23 01:15 -0600

File ▲	statements	missing	excluded	coverage
P1 \ __init__.py	0	0	0	100%
P1 \ reservation_system.py	119	2	0	98%
test \ __init__.py	0	0	0	100%
test \ unit \ __init__.py	0	0	0	100%
test \ unit \ P1 \ __init__.py	0	0	0	100%
test \ unit \ P1 \ reservation_system_test.py	91	1	0	99%
Total	**210**	**3**	**0**	**99%**

coverage.py v7.13.4, created at 2026-02-23 01:15 -0600